

BANK LOAN REPORT

A. PROJECT OVERVIEW

The Bank Loan Analysis project provides a comprehensive view of the bank's lending performance using interactive dashboards and SQL-driven insights. It tracks key metrics such as loan applications, funded amounts, repayments, interest rates, and borrower risk indicators. Loans are classified into Good and Bad categories to assess portfolio quality and identify potential risks. Trend, regional, and borrower-profile analyses help uncover patterns in lending behavior and repayment performance. The project enables data-driven decision-making by giving the bank clear visibility into portfolio health and operational efficiency.

B. DATASET SUMMERY

- Rows: 38000 +
- Columns: 24
- Key Features:
 - Loan & Borrower Identification: (Loan ID, Issue Date, State, Purpose)
 - Financial Details: (Loan Amount, Total Payment Received, Interest Rate, Debt-to-Income Ratio)
 - Loan Status Information: (Current Status, Good Loan / Bad Loan Category)
 - Borrower Profile: (Employment Length, Home Ownership, Loan Grade, Loan Term)
 - Time-Based Attributes: (Month of Issue, Year of Issue - used for MTD & MoM analysis)
 - Segment Analysis Fields: (Loan Purpose, Address State, Term, Home Ownership, Employment Length)

C. BANK LOAN ANALYSIS USING SQL

D. BANK LOAN REPORT | SUMMARY

KPI's:

1. Total Loan Applications

```
select count(id) as Total_Loan_Applications from financial_loan;
```

	total_loan_applications bigint
1	38576

2. Month to date Total Loan applications

```
select count(id) as MTD_Total_Loan_Applications from financial_loan
```

```
where extract(month from issue_date) = 12 and extract(year from issue_date) = 2021;
```

	mtd_total_loan_applications bigint
1	4314

3. previous month to date loan applications

select count(id) as MTD_Total_Loan_Applications from financial_loan

where extract(month from issue_date) = 11 and extract(year from issue_date) = 2021;

	mtd_total_loan_applications bigint
1	4035

4. Total funded amount

select sum(loan_amount) as Total_Funded_Amount from financial_loan;

	total_funded_amount bigint
1	435757075

5. Month to date Total funded amount

select sum(loan_amount) as MTD_Total_Funded_Amount from financial_loan

where extract(month from issue_date) = 12 and extract(year from issue_date) = 2021;

	mtd_total_funded_amount bigint
1	53981425

6. Previous Month to date Total funded amount

select sum(loan_amount) as MTD_Total_Funded_Amount from financial_loan

where extract(month from issue_date) = 11 and extract(year from issue_date) = 2021;

	mtd_total_funded_amount bigint
1	47754825

7. Total Amount Received

select sum(total_payment) as Total_Amount_Received from financial_loan;

	total_amount_received numeric
1	473070933.00

8. Month to date Total amount received

select sum(total_payment) as MTD_Total_Amount_Received from financial_loan

where extract(month from issue_date) = 12 and extract(year from issue_date) = 2021;

	mtd_total_amount_received numeric
1	58074380.00

9. Previous Month to date Total amount received

select sum(total_payment) as MTD_Total_Amount_Received from financial_loan
where extract(month from issue_date) = 11 and extract(year from issue_date) = 2021;

	mtd_total_amount_received numeric
1	50132030.00

10. Average interest rate

select round(avg(int_rate) * 100, 2) as Avg_interest_Rate from financial_loan;

	avg_interest_rate numeric
1	12.05

11. Month to date Average interest rate

select round(avg(int_rate) * 100, 2) as Avg_interest_Rate from financial_loan
where extract(month from issue_date) = 12 and extract(year from issue_date) = 2021;

	avg_interest_rate numeric
1	12.36

12. Previous Month to date Average interest rate

select round(avg(int_rate) * 100, 2) as Avg_interest_Rate from financial_loan
where extract(month from issue_date) = 11 and extract(year from issue_date) = 2021;

	avg_interest_rate numeric
1	11.94

13. Debt To Income ratio

select round(avg(dti) * 100, 2) as Avg_DTI from financial_loan;

	avg_dti numeric
1	13.33

14. Month to date Debt To Income ratio

select round(avg(dti) *100, 2) as Avg_DTI from financial_loan

where extract(month from issue_date) = 12 and extract(year from issue_date) = 2021;

	avg_dti numeric
1	13.67

15. Previous Month to date Debt Income Ratio

select round(avg(dti) *100, 2) as Avg_DTI from financial_loan

where extract(month from issue_date) = 11 and extract(year from issue_date) = 2021;

	avg_dti numeric
1	13.30

16. Good loan application percentage

Select (count(case when loan_status = 'Fully Paid' or loan_status = 'Current' then id end) * 100)

/ count(id) as Good_Loan_Percentage from financial_loan;

	good_loan_percentage bigint
1	86

17. Good loan application

select count(id) as Good_Loan_Applications from financial_loan

where loan_status = 'Fully Paid' or loan_status = 'Current';

	good_loan_applications bigint
1	33243

18. Good Funded Amount

select sum(loan_amount) as Good_Loan_Funded_Amount from financial_loan

where loan_status in ('Fully Paid', 'Current');

	good_loan_funded_amount bigint
1	370224850

19. Good Loan Total Amount

select sum(total_payment) as Good_Loan_Recieved_Amount from financial_loan

where loan_status in ('Fully Paid', 'Current');

	good_loan_recieved_amount numeric
1	435786170.00

20. Bad Loan Applications Percentage

Select (count(case when loan_status = 'Charged Off' then id end) * 100)
/ count(id) as Bad_loan_percentage from financial_loan;

	bad_loan_percentage bigint
1	13

21. Bad loan applications

select count(id) as bad_loan_applications from financial_loan
where loan_status = 'Charged Off';

	bad_loan_applications bigint
1	5333

22. Bad Funded Amount

select sum(loan_amount) as Bad_loan_funded_amount from financial_loan
where loan_status = 'Charged Off';

	bad_loan_funded_amount bigint
1	65532225

23. Bad Loan Total Amount

select sum(total_payment) as bad_loan_recieved_amount from financial_loan
where loan_status = 'Charged Off';

	bad_loan_recieved_amount numeric
1	37284763.00

24. Loan Status Summery

For each loan status, what is the loan count, total amount received, total funded amount, average interest rate, and average DTI?

select loan_status,

count(id) as Total_Loan_Applications, sum(total_payment) as Total_Recieved_Amount,
sum(loan_amount) as Total_Funded_Amount, round(avg(int_rate) * 100, 2) as Interest_rate,
round(avg(dti) * 100, 2) as DTI
from financial_loan
group by loan_status;

	loan_status character varying (30) 🔒	total_loan_applications bigint 🔒	total_recieved_amount numeric 🔒	total_funded_amount bigint 🔒	interest_rate numeric 🔒	dti numeric 🔒
1	Current	1098	24199914.00	18866500	15.10	14.72
2	Fully Paid	32145	411586256.00	351358350	11.64	13.17
3	Charged Off	5333	37284763.00	65532225	13.88	14.00

E. BANK LOAN REPORT | OVERVIEW

25. Monthly Overview

For each month, what are the total loan applications, total funded amount, and total amount received?

select extract (month from issue_date) as Month_Number,
to_char(issue_date, 'Month') as Month_Name, count(id) as Total_Loan_applications,
sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as Total_Recieved_Amount
from financial_loan
group by Month_Number, Month_Name
order by Month_Number;

	month_number numeric 🔒	month_name text 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_recieved_amount numeric 🔒
1	1	January	2332	25031650	27578836.00
2	2	February	2279	24647825	27717745.00
3	3	March	2627	28875700	32264400.00
4	4	April	2755	29800800	32495533.00
5	5	May	2911	31738350	33750523.00
6	6	June	3184	34161475	36164533.00
7	7	July	3366	35813900	38827220.00
8	8	August	3441	38149600	42682218.00
9	9	September	3536	40907725	43983948.00
10	10	October	3796	44893800	49399567.00
11	11	November	4035	47754825	50132030.00
12	12	December	4314	53981425	58074380.00

26. State Overview

For each state, how many loan applications were received, and what are the total funded and total received amounts?

```
select address_state, count(id) as Total_Loan_applications,  
sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as Total_Recieved_Amount  
from financial_loan  
group by address_state  
order by Count(id) desc;
```

	address_state character varying (50) 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_recieved_amount numeric 🔒
1	CA	6894	78484125	83901234.00
2	NY	3701	42077050	46108181.00
3	FL	2773	30046125	31601905.00
4	TX	2664	31236650	34392715.00
5	NJ	1822	21657475	23425159.00
6	IL	1486	17124225	18875941.00
7	PA	1482	15826525	17462908.00
8	VA	1375	15982650	17711443.00
9	GA	1355	15480325	16728040.00
10	MA	1310	15051000	16676279.00

27. Term Overview

By loan term, what are the total applications, funded amount, and collected amount?

```
select term, count(id) as Total_Loan_applications,  
sum(loan_amount) as Total_Funded_Amount, sum(total_payment) as Total_Recieved_Amount  
from financial_loan  
group by term  
order by term desc;
```

	term character varying (20) 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_recieved_amount numeric 🔒
1	60 months	10339	162715850	178361475.00
2	36 months	28237	273041225	294709458.00

28. Employee Length Overview

For each employee length category, how many loan applications were submitted and what are the total funded and collected amounts?

```
select emp_length, count(id) as Total_Loan_applications, sum(loan_amount) as Total_Funded_Amount,  
sum(total_payment) as Total_Recieved_Amount from financial_loan
```

group by emp_length

order by count(id) desc;

	emp_length character varying (20) 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_recieved_amount numeric 🔒
1	10+ years	8870	116115950	125871616.00
2	< 1 year	4575	44210625	47545011.00
3	2 years	4382	44967975	49206961.00
4	3 years	4088	43937850	47551832.00
5	4 years	3428	37600375	40964850.00
6	5 years	3273	36973625	40397571.00
7	1 year	3229	32883125	35498348.00
8	6 years	2228	25612650	27908658.00
9	7 years	1772	20811725	22584136.00
10	8 years	1476	17558950	19025777.00
11	9 years	1255	15084225	16516173.00

29. Loan Purpose Analysis

For each loan purpose, what are the total applications, funded amount, and collected amount?

select purpose, count(id) as Total_Loan_applications, sum(loan_amount) as Total_Funded_Amount,
sum(total_payment) as Total_Recieved_Amount from financial_loan

group by purpose

order by count(id) desc;

	purpose character varying (50) 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_recieved_amount numeric 🔒
1	Debt consolidation	18214	232459675	253801871.00
2	credit card	4998	58885175	65214084.00
3	other	3824	31155750	33289676.00
4	home improvement	2876	33350775	36380930.00
5	major purchase	2110	17251600	18676927.00
6	small business	1776	24123100	23814817.00
7	car	1497	10223575	11324914.00
8	wedding	928	9225800	10266856.00
9	medical	667	5533225	5851372.00
10	moving	559	3748125	3999899.00
11	house	366	4824925	5185538.00
12	vacation	352	1967950	2116738.00
13	educational	315	2161650	2248380.00
14	renewable_energy	94	845750	898931.00

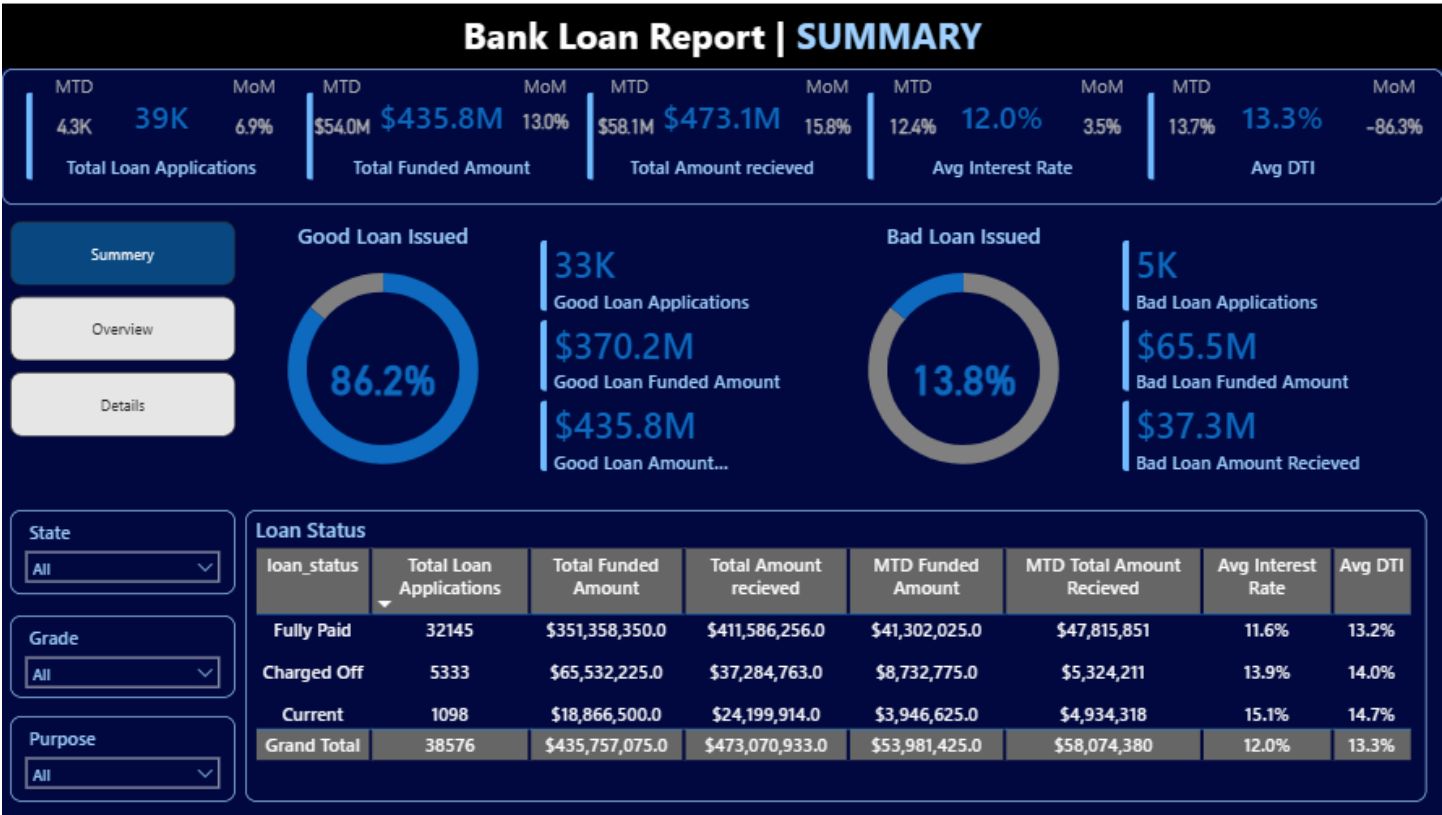
30. Home Ownership Analysis

How do loan applications, funded amounts, and received amounts vary across home ownership categories?

```
select home_ownership, count(id) as Total_Loan_applications, sum(loan_amount) as Total_Funded_Amount,
sum(total_payment) as Total_Recieved_Amount from financial_loan
group by home_ownership
order by count(id) desc;
```

	home_ownership character varying (20)	total_loan_applications bigint	total_funded_amount bigint	total_recieved_amount numeric
1	RENT	18439	185768475	201823056.00
2	MORTGAGE	17198	219329150	238474438.00
3	OWN	2838	29597675	31729129.00
4	OTHER	98	1044975	1025257.00
5	NONE	3	16800	19053.00

F. BANK LOAN DASHBOARD | POWER BI



Bank Loan Report | OVERVIEW

Summary

Overview

Details

Select Measures

Total Loan Applications

State

All

Grade

All

Good vs Bad Loan

All

Total Loan Applications

38.6K

Total Funded Amount

\$435.8M

Total Amount Received

\$473.1M

Average Interest rate

12.0%

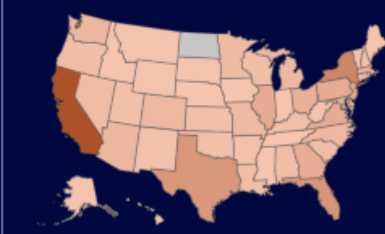
Average DTI

13.3%

Total Loan Applications by Month



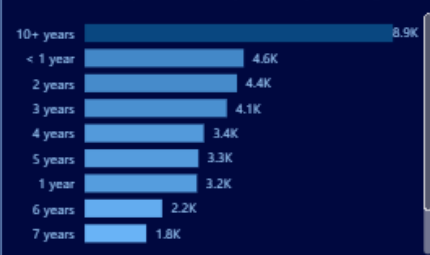
Total Loan Applications by State



Total Loan Applications by term



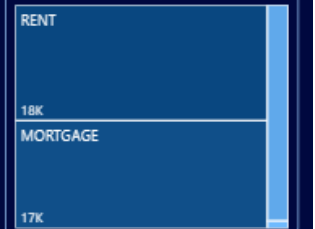
Total Loan Applications by Employee Length



Total Loan Applications by purpose



Total Loan Applications by Home Ownership



Bank Loan Report | DETAILS

Summary

Overview

Details

State

All

Grade

All

Good vs Bad Loan

All

Total Loan Applications

38.6K

Total Funded Amount

\$435.8M

Total Amount Received

\$473.1M

Average Interest rate

12.0%

Average DTI

13.3%

Id	Purpose	Home Ownership		Grade	Sub Grade	Issue Date	Funded Amount	Int Rate	Installment	Amount Received
54734	Debt consolidation	RENT		B	B4	09 August 2021	\$25,000.0	11.9%	\$829	\$29,330.0
55742	credit card	RENT		B	B5	08 May 2021	\$7,000.0	10.7%	\$228	\$8,216.0
57416	Debt consolidation	RENT		C	C3	09 November 2021	\$10,800.0	13.6%	\$367	\$13,208.0
58915	Debt consolidation	RENT		B	B3	08 April 2021	\$7,500.0	10.1%	\$162	\$5,844.0
61419	Debt consolidation	RENT		D	D2	10 February 2021	\$5,600.0	15.0%	\$194	\$6,475.0
62102	Debt consolidation	RENT		B	B1	10 April 2021	\$3,200.0	9.9%	\$103	\$3,414.0
66431	Debt consolidation	RENT		B	B5	09 February 2021	\$2,525.0	12.2%	\$84	\$3,028.0
66943	Debt consolidation	RENT		B	B4	10 August 2021	\$2,800.0	11.5%	\$62	\$3,144.0
68381	Debt consolidation	RENT		A	A5	08 March 2021	\$6,625.0	8.6%	\$210	\$7,542.0
68926	moving	RENT		D	D2	08 August 2021	\$2,300.0	13.2%	\$78	\$2,797.0
69168	major purchase	RENT		E	E4	08 April 2021	\$5,000.0	15.1%	\$174	\$6,251.0
69550	Debt consolidation	RENT		D	D3	10 May 2021	\$11,200.0	15.3%	\$268	\$15,924.0
69924	credit card	RENT		D	D4	08 March 2021	\$10,000.0	13.6%	\$340	\$12,225.0
69990	credit card	RENT		A	A5	10 February 2021	\$15,500.0	7.9%	\$485	\$17,438.0
70348	Debt consolidation	RENT		C	C5	08 March 2021	\$9,600.0	12.3%	\$320	\$11,341.0
88854	house	RENT		A	A2	07 August 2021	\$5,000.0	7.4%	\$155	\$5,174.0
90665	credit card	RENT		C	C1	07 July 2021	\$8,500.0	10.3%	\$275	\$9,913.0

G. BUSINESS RECOMMENDATIONS

- **Implement Risk-Based Pricing** – Adjust interest rates based on employment stability and customer risk profiles.
- **Expand Loan Product Offerings** – Launch specialized debt consolidation and credit card refinancing products.
- **Strengthen Verification Processes** – Enhance employment and income verification to reduce defaults.
- **Focus on High-Performing Segments** – Increase marketing to reliable customer groups in top-performing states.
- **Diversify Loan Terms** – Introduce more flexible loan durations to balance risk and customer needs.
- **Enhance Early Customer Support** – Proactively assist at-risk borrowers to improve repayment rates.